

The empirical recurrence for OEIS A267962: recovering a conjecture that is not written down

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Abstract

OEIS A267962 records “Empirical recurrence of order 60”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 1537$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A267962 is “Number of $n \times 3$ 0..2 arrays with every repeated value in every row and column greater than the previous repeated value.”. It has offset 1 and begins

24, 576, 9750, 157182, 2253828, 30649446, 392792334, 4834114074, ...

The entry states:

Empirical recurrence of order 60 (see link above)

As of the “Last modified” line on the live entry (Jan 22 2016 14:28:29, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1537$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1537$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 3074$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{60} c_i a(n-i),$$

c_1	30	c_{16}	238682468408823	c_{31}	−109484874160268943360	c_{46}	−523260490853916
c_2	−69	c_{17}	1250132604290046	c_{32}	−1013014011093838300416	c_{47}	100733828552938
c_3	−5852	c_{18}	−2671989447018669	c_{33}	−215207851411768130048	c_{48}	187550110586373
c_4	40440	c_{19}	−19280837961286380	c_{34}	1995261633006757565184	c_{49}	263026429973338
c_5	548904	c_{20}	15645491496161964	c_{35}	1230916755693114270720	c_{50}	−34692609180463
c_6	−5093668	c_{21}	215637120675701376	c_{36}	−2688327945045364466688	c_{51}	−14945123408164
c_7	−34059744	c_{22}	30605128651310832	c_{37}	−2750428930994211225600	c_{52}	133485401602208
c_8	361521090	c_{23}	−1730733916914085248	c_{38}	2292930748458203922432	c_{53}	24415481208955
c_9	1634836116	c_{24}	−1518869241011521824	c_{39}	3752323956539776499712	c_{54}	57090252775148
c_{10}	−17169594390	c_{25}	9868592381312431296	c_{40}	−898284746125262389248	c_{55}	−5879638429532
c_{11}	−65658433176	c_{26}	15196763200365412128	c_{41}	−3364774140751530688512	c_{56}	−6153344669397
c_{12}	578744759364	c_{27}	−38959713992701230464	c_{42}	−400051805121861255168	c_{57}	−1549046956792
c_{13}	2213678618856	c_{28}	−90100718058622422144	c_{43}	1976411682330096697344	c_{58}	−204509162766
c_{14}	−14014641594852	c_{29}	98362019513914911744	c_{44}	795416999106892529664	c_{59}	−145135534866
c_{15}	−60053059796928	c_{30}	360955275802684120832	c_{45}	−702006461107506511872	c_{60}	−43980465111

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60 + 4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

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